

An Asymptotic Diagonal Escape Criterion for Quotients

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when K has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Proof Intuition

In a quotient $X = Y/J$, the useful local norming maps are often defined on the ambient space Y , not on the quotient. For example, in a multiplier corona one tests a multiplier on a far-out annulus; adding an element of the ideal changes the annulus, but by a quantity that tends to zero along the tail. The same phenomenon appears in reduced products and Hilbert-module coronas.

The criterion below isolates exactly what the diagonal proof needs. Each center is assigned infinitely many local tests. The assigned tests nearly norm that center. The ideal J vanishes along those tests. Finally, one global quotient unit has prescribed local behavior, chosen to point away from the corresponding local center. These four properties force the quotient unit to avoid every ball $B(\xi_i, \|\xi_i\|)$.

Criterion

Let Y be a normed space, let $J \subseteq Y$ be a closed subspace, and write

$$q : Y \longrightarrow X := Y/J$$

for the quotient map.

Definition 1. *The quotient $X = Y/J$ has the asymptotic diagonal escape property if for every sequence $(y_i)_{i \geq 1} \subset Y$ there are:*

- *a map $\pi : \mathbb{N} \rightarrow \mathbb{N}$ such that every fiber $\pi^{-1}(\{i\})$ is infinite;*
- *normed spaces E_k ;*
- *contractive linear maps $L_k : Y \rightarrow E_k$;*
- *a null sequence $\eta_k \downarrow 0$;*
- *vectors $u_k \in E_k$;*
- *a vector $y \in Y$ with $q(y) \in S_X$;*

such that the following hold.

1. Assigned near-norming.

$$\|L_k y_{\pi(k)}\| \geq \|q(y_{\pi(k)})\| - \eta_k, \quad k \geq 1.$$

2. Ideal vanishing along fibers. *For every $c \in J$ and every $i \geq 1$,*

$$\lim_{\substack{k \rightarrow \infty \\ \pi(k)=i}} \|L_k c\| = 0.$$

3. Asymptotic gluing.

$$\|L_k y - u_k\| \leq \eta_k, \quad k \geq 1.$$

4. Local escape.

$$\|u_k - L_k y_{\pi(k)}\| \geq \|L_k y_{\pi(k)}\| - \eta_k, \quad k \geq 1.$$

Theorem 1. *If $X = Y/J$ has the asymptotic diagonal escape property, then X fails BCP.*

Proof. We use the standard equivalent form of BCP: if X has BCP, then S_X is covered by countably many open balls $B(\xi_i, \|\xi_i\|)$.

Let $(\xi_i)_{i \geq 1} \subset X$ be an arbitrary countable family of centers, and choose representatives $y_i \in Y$ with $q(y_i) = \xi_i$. Apply the asymptotic diagonal escape property to (y_i) , obtaining the data from Definition 1. Put

$$\zeta = q(y) \in S_X.$$

Fix $i \geq 1$. We prove that

$$\|\zeta - \xi_i\| \geq \|\xi_i\|.$$

Let $c \in J$. For each k with $\pi(k) = i$, contractivity gives

$$\|y - y_i - c\| \geq \|L_k(y - y_i - c)\|.$$

Using asymptotic gluing, local escape, and assigned near-norming,

$$\begin{aligned}\|L_k(y - y_i - c)\| &\geq \|u_k - L_k y_i\| - \|L_k y - u_k\| - \|L_k c\| \\ &\geq \|L_k y_i\| - 2\eta_k - \|L_k c\| \\ &\geq \|\xi_i\| - 3\eta_k - \|L_k c\|.\end{aligned}$$

Letting $k \rightarrow \infty$ through the infinite fiber $\pi^{-1}(\{i\})$, and using ideal vanishing, gives

$$\|y - y_i - c\| \geq \|\xi_i\|.$$

Since $c \in J$ was arbitrary,

$$\|\zeta - \xi_i\| = \inf_{c \in J} \|y - y_i - c\| \geq \|\xi_i\|.$$

Thus $\zeta \in S_X$ is outside every open ball

$$B(\xi_i, \|\xi_i\|).$$

The countable family of centers was arbitrary, so X fails BCP. □

How to Produce the Local Escape Vectors

The criterion separates the formal diagonal argument from the local construction. In most applications u_k is obtained by a Hahn–Banach phase or sign choice.

Lemma 1 (local sign escape). *Let $E \neq \{0\}$ be a real or complex normed space and let $v \in E$. There is $u \in S_E$ such that*

$$\|u - v\| \geq \|v\|.$$

Proof. If $v = 0$, choose any $u \in S_E$. If $v \neq 0$, choose $f \in S_{E^*}$ with $f(v) = \|v\|$. Choose $w \in S_E$ with $|f(w)| > 0$, and multiply w by a scalar of modulus one in the complex case, or change sign in the real case, so that $u \in S_E$ satisfies $\operatorname{Re} f(u) \leq 0$. Then

$$\|u - v\| \geq \operatorname{Re} f(v - u) \geq \|v\|.$$

□

Relation to the Exact Criterion

The exact adaptive diagonal criterion from the previous packet is the special case $J = \{0\}$ where the local maps L_k are genuine quotient maps $R_k : X \rightarrow E_k$, the gluing is exact, and $\eta_k = 0$. In particular, if one has a linear isometry

$$\Gamma : \ell_\infty(E_k) \rightarrow X, \quad R_k \Gamma((u_j)) = u_k,$$

and the assigned coordinates nearly norm the centers, then choosing local sign escapes u_k and setting $y = \Gamma((u_k))$ verifies Definition 1.

How the Earlier Diagonal Packets Fit

This section is a guide for applying the criterion; it is not a replacement for the full proofs in the earlier packets.

Sigma-Unital Multiplier Coronas

For a nonunital σ -unital C^* -algebra A , take

$$Y = \mathcal{M}(A), \quad J = A, \quad X = \mathcal{M}(A)/A.$$

With an increasing approximate unit (e_n) satisfying $e_m e_n = e_n$ for $m \geq n$, the local tests are

$$L_k(T) = Tg_k, \quad g_k = e_{m_k} - e_{n_k},$$

where the annuli g_k are chosen far out. Finite annulus recovery gives assigned near-norming:

$$\|T_{\pi(k)}g_k\| \approx \|q(T_{\pi(k)})\|.$$

For $C \in A$, tail convergence gives

$$\|Cg_k\| \rightarrow 0,$$

which is ideal vanishing. The local escape vectors are phased annuli $u_k = \lambda_k g_k$, chosen by the sign trick against $T_{\pi(k)}g_k$. The global multiplier

$$z = \sum_k \lambda_k h_k$$

over larger orthogonal annuli satisfies $zg_k = u_k$, and an auxiliary subsequence of annuli verifies $\|q(z)\| = 1$. These are exactly the ingredients of Definition 1.

Hilbert-Module Coronas

For a Hilbert module E with a BCP block system, take

$$Y = \mathcal{L}(E), \quad J = \mathcal{K}(E), \quad X = \mathcal{L}(E)/\mathcal{K}(E).$$

The local tests are

$$L_k(T) = Te_{F_k},$$

where e_{F_k} is a fresh finite block projection. Tail norm recovery gives assigned near-norming, compact-vanishing gives ideal vanishing, fresh range escape supplies local $u_k = v_k$, and strict gluing supplies the global operator V with $Ve_{F_k} = v_k$. The previous Hilbert-module packet is therefore an instance of the quotient criterion.

Reduced Products

For reduced products

$$X = \ell_\infty(E_n)/c_{\mathcal{I}}(E_n),$$

the ambient space is $Y = \ell_\infty(E_n)$ and $J = c_{\mathcal{I}}(E_n)$. Given centers, the proof chooses disjoint $\mathcal{I}$ -positive sets T_k on which the assigned center is nearly normed. The local maps are restrictions

$$L_k((a_n)) = (a_n)_{n \in T_k} + c_{\mathcal{I}|T_k}(E_n : n \in T_k)$$

viewed in the local reduced quotient

$$\ell_\infty(E_n : n \in T_k)/c_{\mathcal{I}|T_k}(E_n : n \in T_k).$$

Thus elements of J vanish exactly under every L_k . The glued sequence is built by putting the locally escaping representatives on the disjoint sets T_k and zero elsewhere; the reduced-product norm formula checks that the resulting quotient class is a unit. This is the same asymptotic diagonal pattern in coordinate form.

Verification and Search Bounds

The run indexes were searched for arXiv:2311.14257 together with approximate adaptive diagonal, asymptotic diagonal, quotient diagonal, tail error, sigma-unital corona, reduced product, Hilbert-module block, and atomless von Neumann keywords. Existing packets contained the exact adaptive criterion and several concrete asymptotic diagonal arguments, but no quotient-level criterion isolating the common proof.

The proof of Theorem 1 is formal once the definition is verified in an application. The application sketches above are calibrated against the already promoted concrete packets.

Human Review Focus

The main point to review is Definition 1: it should be checked against future applications to ensure that the hard local construction is not hidden inside the definition. The theorem itself is a short quotient-norm diagonal estimate.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.